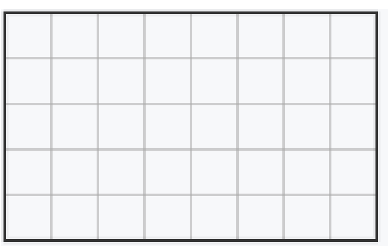
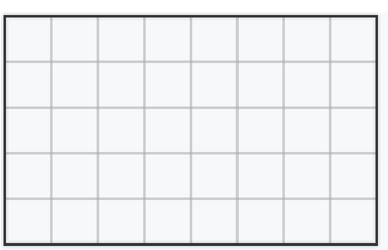
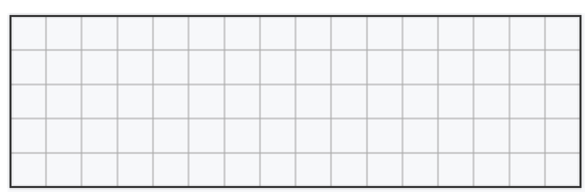
Lesson Objective

Analyze different rectangles and reason about their area.

Materials: Centimeter grid paper, personal white board

1. Find the area of two equal rectangles and add to find the total area. Here are 2 equal rectangles. Label the side lengths. Write an equation to find the area of each rectangle. Then write an equation to show the total area.  	Show students the two equal rectangles on the centimeter grid. Point to one rectangle and ask students to count the side lengths and label them on the rectangle. Ask students how they can find the area of one rectangle. Have students write a multiplication equation and find the area. Ask students for the area of the other rectangle and how they know. Guide students to see that because the rectangles are equal, the area is the same. Ask students to write an addition equation to find the total area of the two rectangles. Confirm the total area is 80 square units.

Answer: Side lengths: 5 units and 8 units for each rectangle; 5 × 8 = 40 sq units; 5 × 8 = 40 sq units; 40 + 40 = 80 sq units	
2. Combine two equal rectangles into a longer rectangle and find its area. Push the two equal rectangles from Problem 1 together along the matching side to make one longer rectangle. Draw the new rectangle and label the side lengths. What is the area of the longer rectangle? *Answer: Side lengths: 5 units and 16 units; Area = 80 sq units*	Have students visualize sliding the two equal rectangles next to each other so the matching sides line up. Ask students what the new side lengths are. Guide them to see that one side is still 5 units, and the longer side is 8 + 8, or 16 units. Ask students to predict the area of the longer rectangle before computing. Prompt students to explain why the area must equal the sum of the two smaller areas from Problem 1. Point out that the total area does not change when the rectangles are joined, because we are just rearranging the same square units. The longer rectangle has an area of 80 square units, the same as 40 + 40 from Problem 1. 

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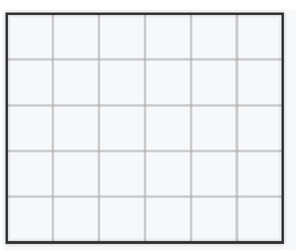
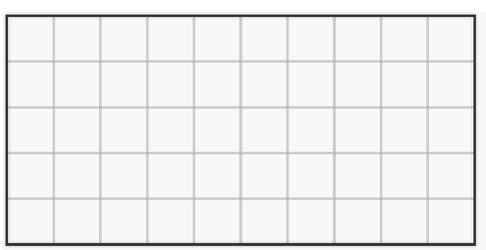
3. Find the area of a longer rectangle by adding the areas of two unequal rectangles joined together.

Here are 2 rectangles that share a side length of 5 units. One rectangle is 5 units by 6 units, and the other is 5 units by 10 units. Find the area of each rectangle. Then push them together along the matching side to make one longer rectangle. What is the total area of the longer rectangle?





Answer: 5 × 6 = 30 sq units; 5 × 10 = 50 sq units; 30 + 50 = 80 sq units

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Have students label the side lengths on each rectangle and write a multiplication equation to find the area of each.

As students are ready, have them draw the longer rectangle formed by joining the two rectangles along the matching side of 5 units, and label its side lengths as 5 units and 16 units.

Have students find the total area by adding the two smaller areas. Ask students how they know the total area of the longer rectangle is 80 square units without multiplying 5 × 16 directly.

Confirm that adding the areas of the two smaller rectangles gives the total area of the longer rectangle, even when the two rectangles are not equal.

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Monitor For	**Instructional Moves**

<table>
<tr><td>• Do students label both side lengths on each rectangle before writing a multiplication equation in Problem 1?

• Do students multiply the side lengths to find area, or do they count individual squares one by one?

• In Problem 2, do students recognize that the shared side stays the same (5 units) and the other side becomes the sum (8 + 8 = 16 units)?

• Do students see that the total area of the longer rectangle equals the sum of the two smaller areas, without recomputing?</td><td>• Before students write a multiplication equation, have them point to and say each side length out loud, such as "5 units by 8 units."

• If a student counts squares one by one to find area, prompt them with, "What multiplication equation matches this rectangle?" to reinforce side length × side length.

• When joining rectangles in Problems 2 and 3, have students trace the matching side with a finger and ask, "Which side stays the same? Which side gets longer?"

• If a student tries to multiply 5 × 16 in Problem 3 and gets stuck, remind them they already found the two smaller areas and can add them to get the total.</td></tr>
<tr><td colspan="2">Reflection Questions

• How did you find the area of each rectangle in Problems 1 and 3? Answer: I multiplied the two side lengths.

• When you pushed the two rectangles together in Problem 2, why was the total area still 80 square units? Answer: We didn't add or take away any squares. We just moved them next to each other, so the total stays the same.

• In Problem 3, how did adding the two smaller areas help you find the area of the longer rectangle? Answer: The longer rectangle is just the two smaller rectangles joined together, so its area is the sum of their areas: 30 + 50 = 80 square units.</td></tr>
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• What stays the same and what changes when you join two rectangles that share a side length?

Answer: The shared side length stays the same. The other side gets longer because it's the sum of the two non-matching sides. The total area is the sum of the two smaller areas.